Parametric APDL-Based Horizontal Tailplane Structural Analysis and Optimization

Group Members

Archawin Chindarattanaporn 67130043
Aryan Sajith 67130044
Thanatchai Khunpiluke 67130036
Sirasak Permsup 67130029
Peeradol Singsamran 67130061

Introduction

The horizontal tailplane (HTP) is a critical load-bearing component of any aircraft, responsible for pitch stability and control. Designing it efficiently requires balancing structural strength, weight, and manufacturability. These concerns multiply as the aviation industry moves toward hybrid-electric and hydrogen-powered configurations with new loading environments.

Traditional finite element analysis workflows for aircraft structures rely on manually constructed models that are time-consuming to modify and difficult to iterate on during early design phases. This limits the number of configurations an engineer can evaluate before committing to a design.

ANSYS Mechanical APDL (ANSYS Parametric Design Language) offers a scripting-based alternative: a fully parametric model where geometry dimensions, rib/spar counts, stringer layout, material properties, and load cases are all controlled by variables. Changing a single parameter regenerates the entire model — geometry, mesh, boundary conditions, and analysis — without manual intervention.

This project proposes developing such a parametric APDL script for an HTP structure, coupled with an external load pre-processing step and a structural optimization loop. The approach follows the methodology demonstrated by Palumbo et al. (2026) in the HERFUSE project, adapted for our coursework scope and the ANSYS tools available in our laboratory

Laboratory Aim

The primary objectives of this project are:

1. Develop a fully parametric ANSYS APDL script that generates the geometry, mesh, material definitions, and boundary conditions for a horizontal tailplane structure. Key design variables — span, chord, number of ribs, number of spars, stringer count, and skin thickness — must be adjustable without manual model editing.
2. Implement a load pre-processing module that converts internal sizing loads (shear forces and bending moments) into external distributed loads suitable for the finite element model, using a least-norm formulation to enforce global equilibrium.
3. Integrate the parametric model with a structural optimization loop to evaluate alternative HTP configurations and identify designs that minimise mass while satisfying strength and stiffness requirements.
4. Verify the model through modal analysis (free-free rigid-body modes), unit displacement checks, and gravity load equilibrium tests to confirm numerical accuracy and physical consistency.
5. Document the workflow so that future design iterations — including changes to airfoil shape, material system, or load cases — require only parameter updates rather than model reconstruction.

Related Theories

Finite Element Method (FEM) for thin-walled structures. Aircraft tailplane structures consist of skins, ribs, spars, and stringers — thin-walled members whose behavior is governed by plate bending and membrane stress. In ANSYS APDL, SHELL181 (4-node structural shell) models ribs, spars, and skins, while BEAM188 (2-node Timoshenko beam) represents stringers with parametrically defined cross-sections. The choice of equivalent isotropic properties derived from a composite laminate stacking sequence allows global stiffness and mass to be captured without full ply-by-ply modelling, which is appropriate for preliminary design.

Parametric modelling in APDL. ANSYS Parametric Design Language allows geometry, mesh density, material cards, and load steps to be driven by scalar parameters and *DO loops. A single script regenerates the entire model when any input changes, eliminating manual rebuild. This is the foundation that enables coupling to an external optimiser.

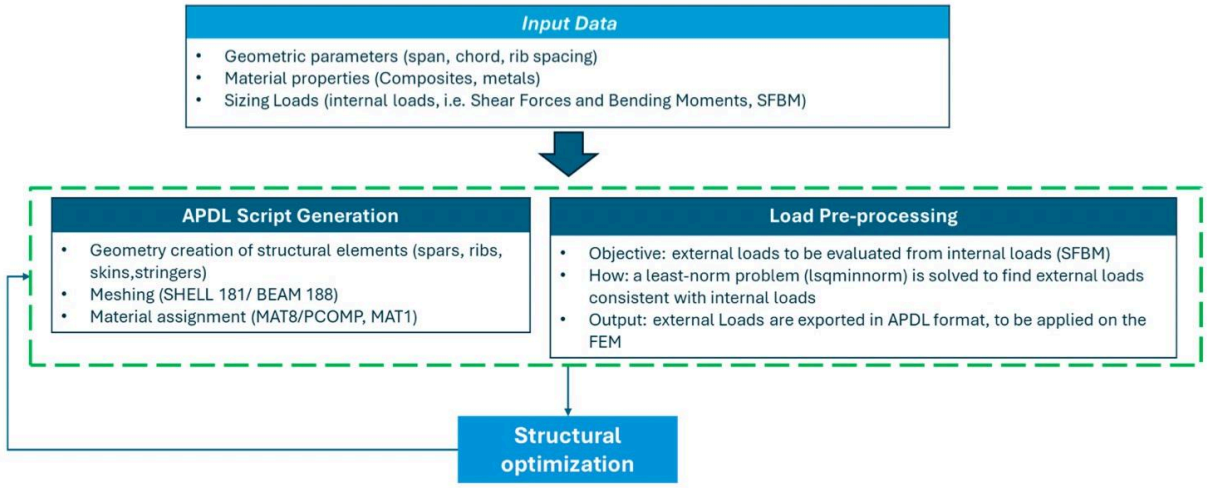
Shear force and bending moment (SFBM) load reconstruction. Aerodynamic loads on the HTP are specified as internal shear forces and bending moments at discrete spanwise stations. To apply them in a 3-D FEM, an inverse problem is solved: find the set of external nodal forces whose resulting internal loads match the target SFBM distribution. The SFBM procedure uses polynomial basis functions (quadratic and cubic) and solves a least-norm problem (via lsqminnorm in MATLAB) to enforce global equilibrium while distributing forces realistically along the span.

Structural optimization. With the parametric model and automated load application in place, a design-of-experiments or evolutionary algorithm loop can sweep structural parameters such as

rib count, spar count, stringer spacing, skin thickness, and evaluate each configuration's mass, maximum stress, and tip displacement. The strength ratio (allowable stress / applied stress > 1) serves as the primary feasibility constraint.

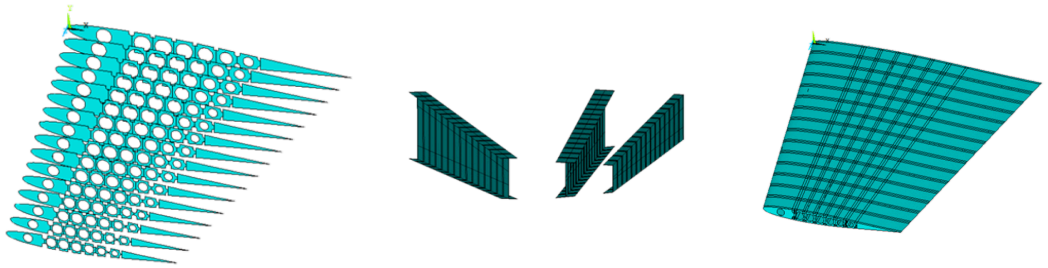
Modal analysis and model verification. A free-free modal analysis must return six near-zero-frequency rigid-body modes before any flexible modes appear; their absence indicates an incomplete stiffness path (disconnected elements or missing constraints). Unit displacement tests confirm correct assembly of the global stiffness matrix, and 1 g gravity checks verify that reaction forces equal the model's total weight.

Methodology



The project follows a three-stage workflow adapted from Palumbo et al. (2026):

Stage 1: Input Data. The user defines geometric parameters (span, root/tip chord, rib spacing, dihedral angle, thickness-to-chord ratio), material properties (equivalent isotropic properties for composite skins, aluminium for metallic parts), and internal sizing loads from the SFBM procedure. All values enter the script as APDL parameters.



[(a) Transverse Ribs (b) Longitudinal Spars (c) Full Assembly and Meshing]

Stage 2a: APDL Script Generation. The script builds the HTP geometry using *DO loops that iterate along the wingspan, creating ribs, spars, skin surfaces, and stringer lines. Cut operations model rib lightening holes. SHELL181 elements discretise ribs, spars, and skins; BEAM188 elements model stringers. Material cards (MAT8/PCOMP for composites, MAT1 for metals) are assigned parametrically. Mesh density is controlled by a single ESIZE variable. Boundary conditions are applied at the HTP–VTP interface root.

Stage 2b: Load Pre-processing. In parallel, a MATLAB-based module (SFBM-HERFUSE) reconstructs external spanwise loads from the target internal shear forces and bending moments. It uses polynomial basis functions and a least-norm solver to find the minimal-norm set of external forces consistent with the specified SFBM distribution. The output is an APDL-formatted load file applied directly to the FEM.

Stage 3a: Configuration Setup. The user defines specific HTP design scenarios to evaluate within the APDL script. To systematically assess weight reduction and structural efficiency, three distinct configurations will be established:

- **Configuration A (Baseline):** Replicates the original literature's reference model, featuring 14 transverse ribs, 3 longitudinal spars, 8 stringers, and a baseline equivalent isotropic skin thickness.
- **Configuration B (Moderate Reduction):** Reduces the internal framework to 9 ribs and 2 spars, while slightly increasing the skin thickness parameter to compensate for the redistributed aerodynamic loads.
- **Configuration C (Aggressive Reduction):** Further simplifies the structure to 7 ribs, 2 spars, and 6 stringers, aiming to find the absolute minimum structural mass that can still withstand the applied bending and shear stresses.

Stage 3b: Model Verification. Before applying aerodynamic loads, each configuration undergoes preliminary checks to ensure numerical accuracy and physical consistency. A free-free modal analysis must yield six near-zero-frequency rigid-body modes, confirming proper element connectivity. A 1 g gravity static analysis is then performed to verify that the structural mass properly reflects the reaction forces at the root.

Stage 3c: Static Analysis. The reconstructed SFBM aerodynamic loads are applied to the verified models. The APDL script solves for the structural response, extracting the maximum Von Mises stresses (primarily bending and shear stresses on the skins and webs) and tip deflections.

Stage 3d: Evaluation and Optimization. Results from all configurations are extracted and compared. The optimal HTP design is identified by finding the configuration that minimizes overall structural mass while strictly maintaining a strength Factor of Safety (FS) greater than 1.5 (aligning with CS-VLA standards). *Furthermore, while the current scope primarily focuses on static strength and mass optimization, the developed fully parametric script provides a robust foundation for future structural stability evaluations, such as buckling analysis.*

Tools required: ANSYS Mechanical APDL (2023 R1 or later), MATLAB (for SFBM load pre-processing).

Conclusion

This project will produce a reusable, parametric APDL workflow for horizontal tailplane structural analysis that can be iterated rapidly without manual model reconstruction. By coupling automated geometry and mesh generation with least-norm load reconstruction, the workflow enables a systematic evaluation of weight-reduced HTP configurations within the laboratory timeframe.

The expected deliverables are: a working APDL script with parametric inputs for key structural variables (such as rib, spar, and stringer counts), a MATLAB load pre-processing module, model verification results (free-free modal and 1 g gravity checks), and a comparative static analysis evaluating the mass and strength ratio of the defined configurations.

Limitations of the current scope include the exclusion of structural stability (buckling) evaluations, the use of equivalent isotropic material properties rather than full ply-by-ply composite modelling, and the reliance on a simplified airfoil profile. These simplifications are acceptable at the preliminary design level and are consistent with the scope of this coursework. Ultimately, the methodology establishes a robust foundation that can be easily extended in future work to incorporate eigenvalue buckling analysis, composite layup definitions, aeroelastic coupling, and broader multidisciplinary design optimization frameworks.

References

Palumbo, C. et al. A Parametric APDL-Based Workflow for Efficient Structural Design of a Civil Aircraft Tailplane. Eng. Proc. 2026, 133, 15.